TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

Team Details

THINK. BUILD. CREATE. WIN

TELIPORT
Script Your Future
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Test a battery before you build it

VOLT-FORGE connects every battery design decision to what it physically changes

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PROBLEM UNDERSTANDING

Cells never share the load evenly

Say you need a Raspberry Pi 5 gateway to survive a **10-minute power cut at 25 W**. Four cells in parallel sounds easy, until the cell closest to the output takes more current than the rest, runs hotter and sets the limit for the whole pack.

DESIRABILITY

You shouldn't need a lab to learn that

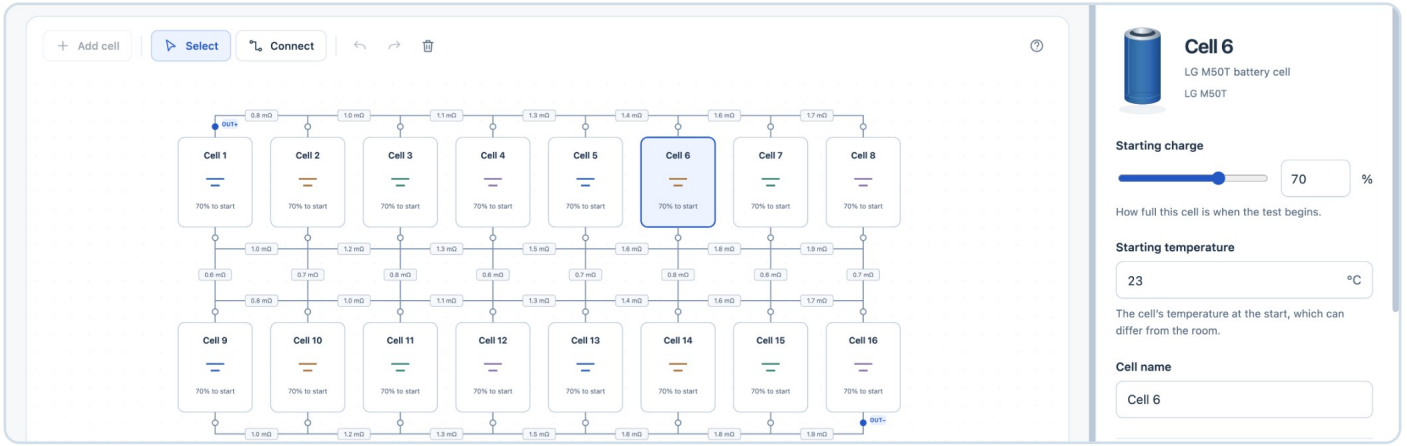
Right now you find out by buying parts and hoping. A student can't afford that, and an engineer can't wait for it. Here you wire the circuit on screen, run it and see which cell struggles **before spending a rupee**.

TECHNICAL FEASIBILITY

So how few cells can do this job?

2 LG M50T cells passed all 33 required cases, even with a weak cell in either position. The busiest cell peaked at 5.33 A against its 7.275 A limit, and one cell alone was refused outright. All of it runs on a laptop.

PROTOTYPE & EVIDENCE A real 16-cell circuit, then two circuits compared under the same 16 A load



16-cell backup module



12-cell backup module



Result	First test	Second test	Difference
Cells	16	12	−4.00
Listed cost, priced parts	₹19,296 + US$52	₹14,472 + US$52	
Cell mass	1,120 g	840 g	−280.00 g
Margin above voltage target	0.76 V	0.67 V	−0.10 V
Warmest cell	25.32 °C	25.92 °C	+0.60 °C
Largest cell current	3.61 A	3.86 A	+0.25 A

Result	First test	Second test	Difference
Energy delivered	19.93 Wh	19.76 Wh	−0.17 Wh
Energy received	0.00 Wh	0.00 Wh	0.00 Wh
Net energy output	19.93 Wh	19.76 Wh	−0.17 Wh
Lowest battery voltage	7.36 V	7.27 V	−0.10 V
Temperature difference	1.44 °C	1.24 °C	−0.19 °C
Test duration	600.00 s	600.00 s	0.00 s

ENGINEERING QUALITY

Now, why should you trust that?

Heat is just electricity that never reached the load. So every run has to balance: energy out of the cells equals energy delivered plus heat. If it doesn't, the result gets flagged. Cooling varies by rig, so temperature stays a **range**.

INNOVATION

You don't have to guess the answer

You could try every cell count by hand. Or press **Find the smallest design**: it rebuilds the circuit at each size, tests them all and shows what survives. For the Pi, that meant 2 cells, not 4, at half the cell cost.

BUSINESS VIABILITY



41.3 lakh TAM, engineering students

15.6 lakh SAM, electrical, electronics, mechanical

130 SOM, colleges, 5% assumed

₹35 crore

to build all of it, from a measured dataset to the full app. 100 person-years, about 2% of AspenTech's yearly R&D.

4,006 what-if runs

Weak cells, bad links, poor cooling: we tried to break it first.

42 overloads caught

Cells pushed past their limit, stopped before anyone bought a part.

10 minutes in 3 seconds

Even the 16-cell module, on an 8 GB laptop.

Everything here is simulated. Next we bench-test real cells and put it in front of students and engineers.